

End-Semester Exam

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For the solution of the differential equation

$$y'' + 8y' + (\lambda + 16)y = 0,$$

the Auxiliary Equation is: $m^2 + 8m + (\lambda + 16) = 0$

$$\Rightarrow m = \frac{-8 \pm \sqrt{8^2 - 4(\lambda + 16)}}{2}$$

$$= \frac{-8 \pm \sqrt{-4\lambda}}{2} = \boxed{-4 \pm \sqrt{-\lambda}}$$

+2

Case 1 If $\lambda < 0$, let $\lambda = -\alpha^2$ where $\alpha > 0$

Then $m = -4 \pm \alpha$

So, the general solution is

$$y(x) = c_1 e^{(-4+\alpha)x} + c_2 e^{(-4-\alpha)x}$$

Applying the boundary condition

$$y(0) = 0 \Rightarrow c_1 + c_2 = 0 \quad \text{--- (1)}$$

and $y(\pi) = 0 \Rightarrow c_1 e^{(-4+\alpha)\pi} + c_2 e^{(-4-\alpha)\pi} = 0$

$$\Rightarrow e^{-4\pi} (c_1 e^{\alpha\pi} + c_2 e^{-\alpha\pi}) = 0$$

$$\Rightarrow c_1 e^{\alpha\pi} + c_2 e^{-\alpha\pi} = 0 \quad \text{--- (2)}$$

Solving (1) and (2) we get $c_1 = c_2 = 0$

$\Rightarrow y(x) = 0$ and no nontrivial solution exists

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Case 2 If $\lambda = 0$, then $m = -4, -4$ (two roots are equal)

and the general solution is

$$y(x) = (c_1 + c_2 x) e^{-4x}$$

Now $y(0) = 0 \Rightarrow c_1 = 0$

$y(\pi) = 0 \Rightarrow c_2 \pi e^{-4\pi} = 0 \Rightarrow c_2 = 0$

Hence $y(x) \equiv 0$ and no nontrivial solution exists

Case 3: Let $\lambda > 0$ Let $\sqrt{\lambda} = \mu$ ($\mu > 0$)

Then $m = -4 \pm i\mu$

So, the general solution is

$$y(x) = e^{-4x} [c_1 \cos(\mu x) + c_2 \sin(\mu x)]$$

Now $y(0) = 0 \Rightarrow c_1 = 0$

Then $y(x) = c_2 e^{-4x} \sin(\mu x)$

Now $y(\pi) = 0 \Rightarrow c_2 e^{-4\pi} \sin(\mu \pi) = 0$

For non trivial solution $c_2 \neq 0$

and $\sin(\mu \pi) = 0 \Rightarrow \mu \pi = n\pi \Rightarrow \mu = n$
for $n = 1, 2, \dots$

So, the eigen values are $\lambda_n = n^2$ for $n = 1, 2, \dots$

and the corresponding eigen functions are $y_n(x) = e^{-4x} \sin(nx)$

+4

Now to ~~the~~ show orthogonality,
we note that the weight function

$$r(x) = p \cdot h \quad \text{where} \quad p = e^{\int 8 dx} = e^{8x} \\ \text{and } h = 1$$

Hence the weight function $r(x) = e^{8x} \cdot 1 = e^{8x}$

Now
$$\int_0^{\pi} y_n(x) y_m(x) e^{8x} dx$$

$$= \int_0^{\pi} e^{-4x} \sin(nx) e^{-4x} \sin(mx) e^{8x} dx$$

$$= \int_0^{\pi} \sin(nx) \sin(mx) dx$$

$$= \frac{1}{2} \int_0^{\pi} [\cos\{(n-m)x\} - \cos\{(n+m)x\}] dx$$

If $m \neq n$

the above
$$= \frac{1}{2} \left[\frac{\sin(n-m)x}{(n-m)} - \frac{\sin(n+m)x}{(n+m)} \right]_0^{\pi}$$

$$= 0$$

If $m=n$, the above
$$= \frac{1}{2} \int_0^{\pi} [1 - \cos(2nx)] dx$$

$$= \frac{1}{2} \left[x - \frac{\sin(2nx)}{2n} \right]_0^{\pi} = \frac{\pi}{2} \neq 0$$

So, the orthogonality of eigen functions has been shown.

$$\textcircled{2} \quad \left. \begin{aligned} x_1' &= x_1 - x_2 + e^{2t} \\ x_2' &= -4x_1 + x_2 + t \end{aligned} \right\} \quad x_1(0) = 0, x_2(0) = 1.$$

The above system can be written as

$$X' = A X + g(t)$$

where $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $A = \begin{bmatrix} 1 & -1 \\ -4 & 1 \end{bmatrix}$, $g(t) = \begin{pmatrix} e^{2t} \\ t \end{pmatrix}$

Now for the general solution for the homogeneous system $X' = AX$,

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -1 \\ -4 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 - (-4)(-1)$$

$$= 1 - 2\lambda + \lambda^2 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$$

$\times 2$ So, the eigen values are 3, -1.

Now for ~~the~~ eigen vector corresponding to the eigen value $\lambda = 3$,

$$\begin{bmatrix} 1-3 & -1 \\ -4 & 1-3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} -2x - y = 0 \\ -4x - 2y = 0 \end{cases} \Rightarrow y = -2x$$

Taking $x = 1$, we can

$\times 1$ See that $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ is an eigen vector corresponding to $\lambda = 3$

(5)

Now for eigen vector corresponding to the eigen value $\lambda = -1$,

$$\begin{bmatrix} 1 - (-1) & -1 \\ -1 & 1 - (-1) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} 2x - y = 0 \\ -4x + 2y = 0 \end{cases} \Rightarrow y = 2x$$

Taking $x=1$, we can see that

(+1) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigen vector corresponding to the eigen value $\lambda = -1$

So, the solution of the homogeneous system

is:
$$Y_h = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-t}$$

(+1)
$$= \begin{bmatrix} e^{3t} & e^{-t} \\ -2e^{3t} & 2e^{-t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = Y c$$

where
$$Y = \begin{bmatrix} e^{3t} & e^{-t} \\ -2e^{3t} & 2e^{-t} \end{bmatrix} \quad \text{and} \quad c = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

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Now for variation of parameter

$$\begin{aligned} Y^{-1} &= \frac{1}{[2e^{2t} - (-2)e^{2t}]} \begin{bmatrix} 2e^{-t} & -e^{-t} \\ 2e^{3t} & e^{3t} \end{bmatrix} \\ &= \frac{1}{4e^{2t}} \begin{bmatrix} 2e^{-t} & -e^{-t} \\ 2e^{3t} & e^{3t} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2e^{-3t} & -e^{-3t} \\ 2e^t & e^t \end{bmatrix} \end{aligned}$$

$$\text{So, } u' = Y^{-1}g = \frac{1}{4} \begin{bmatrix} 2e^{-3t} & -e^{-3t} \\ 2e^t & e^t \end{bmatrix} \begin{bmatrix} e^{2t} \\ t \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 2e^{-t} - te^{-3t} \\ 2e^{3t} + te^t \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2}e^{-t} - \frac{t}{4}e^{-3t} \\ \frac{1}{2}e^{3t} + \frac{t}{4}e^t \end{bmatrix}$$

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Integrating

$$U = \int \begin{pmatrix} \frac{1}{2} e^{-t} - \frac{t}{4} e^{-3t} \\ \frac{1}{2} e^{3t} + \frac{t}{4} e^t \end{pmatrix} dt$$

$$= \begin{bmatrix} -\frac{1}{2} e^{-t} + \frac{t}{12} e^{-3t} + \frac{1}{36} e^{-3t} \\ \frac{1}{6} e^{3t} + \frac{t}{4} e^t - \frac{1}{4} e^t \end{bmatrix} \quad (+2)$$

$$\text{Then } X_p = YU = \begin{bmatrix} e^{3t} & e^{-t} \\ -2e^{3t} & 2e^{-t} \end{bmatrix} \begin{bmatrix} -\frac{1}{2} e^{-t} + \frac{t}{12} e^{-3t} + \frac{1}{36} e^{-3t} \\ \frac{1}{6} e^{3t} + \frac{t}{4} e^t - \frac{1}{4} e^t \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{2} e^{2t} + \frac{t}{12} + \frac{1}{36} + \frac{1}{6} e^{2t} + \frac{t}{4} - \frac{1}{4} \\ e^{2t} - \frac{t}{6} - \frac{1}{18} + \frac{1}{3} e^{2t} + \frac{t}{2} - \frac{1}{2} \end{bmatrix}$$

(+1)

$$= \begin{bmatrix} -\frac{1}{3} e^{2t} + \frac{t}{3} - \frac{2}{9} \\ \frac{4}{3} e^{2t} + \frac{t}{3} - \frac{5}{9} \end{bmatrix}$$

So, the general solution of the system is

$$X = X_h + X_p$$

$$= c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-t} + \begin{bmatrix} -\frac{1}{3}e^{2t} + \frac{t}{3} - \frac{2}{9} \\ \frac{4}{3}e^{2t} + \frac{t}{3} - \frac{5}{9} \end{bmatrix}$$

$$\left. \begin{aligned} \text{Now } x_1(0) = 0 &\Rightarrow c_1 + c_2 - \frac{2}{9} - \frac{1}{3} = 0 \\ x_2(0) = 1 &\Rightarrow -2c_1 + 2c_2 - \frac{5}{9} + \frac{4}{3} = 1 \end{aligned} \right\}$$

$$\Rightarrow c_1 + c_2 = \frac{5}{9} \quad \text{and} \quad -2c_1 + 2c_2 = \frac{2}{9}$$

$$\Rightarrow \left. \begin{aligned} c_1 + c_2 &= \frac{5}{9} \\ -c_1 + c_2 &= \frac{1}{9} \end{aligned} \right\} \Rightarrow 2c_2 = \frac{6}{9} = \frac{2}{3} \Rightarrow c_2 = \boxed{\frac{1}{3}}$$

$$\text{and } c_1 = \frac{5}{9} - \frac{1}{3} = \boxed{\frac{2}{9}}$$

Therefore ~~$X = \frac{2}{9} \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{3t} + \frac{1}{3} \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t} + \begin{pmatrix} -\frac{1}{3}e^{2t} + \frac{t}{3} - \frac{2}{9} \\ \frac{4}{3}e^{2t} + \frac{t}{3} - \frac{5}{9} \end{pmatrix}$~~

OR

$$x_1(t) = \frac{2}{9}e^{3t} + \frac{1}{3}e^{-t} - \frac{1}{3}e^{2t} + \frac{t}{3} - \frac{2}{9}$$

$$x_2(t) = -\frac{4}{9}e^{3t} + \frac{2}{3}e^{-t} + \frac{4}{3}e^{2t} + \frac{t}{3} - \frac{5}{9}$$

Note:

They can write either way.

(9)

(3) (a) $U_{xx} + 5U_{xy} + 4U_{yy} = 0$

Comparing with $AU_{xx} + 2BU_{xy} + CU_{yy} = 0$

we have $A = 1, B = \frac{5}{2}, C = 4$

(+2) Now $AC - B^2 = (1)(4) - \left(\frac{5}{2}\right)^2 = 4 - \frac{25}{4}$
 $= -\frac{9}{4} < 0$

Hence the equation is hyperbolic equation

(b) The characteristic equation is

$y'^2 - 2 \cdot \frac{5}{2} y' + 4 = 0$

$\Rightarrow y'^2 - 5y' + 4 = 0$

$\Rightarrow (y' - 1)(y' - 4) = 0$

Now $y' - 1 = 0 \Rightarrow y' = 1 \Rightarrow y = x + c_1' \Rightarrow x - y = c_1$
 $\Rightarrow \Phi(x, y) = x - y = c_1$

and $y' - 4 = 0 \Rightarrow y' = 4 \Rightarrow y = 4x + c_2' \Rightarrow 4x - y = c_2$
 $\Rightarrow \Psi(x, y) = 4x - y = c_2$

So, the change of variable will be

$v = \Phi(x, y) = x - y$

$w = \Psi(x, y) = 4x - y$

(+2)

Now $U_x = U_v v_x + U_\omega \omega_x = U_v (1) + U_\omega (4)$
 $= U_v + 4U_\omega$

(+1)
$$\begin{aligned} U_{xx} &= (U_v + 4U_\omega)_v v_x + (U_v + 4U_\omega)_\omega \omega_x \\ &= U_{vv} + 4U_{\omega v} + 4U_{v\omega} + 16U_{\omega\omega} \\ &= U_{vv} + 8U_{v\omega} + 16U_{\omega\omega} \end{aligned}$$

(+1)
$$\begin{aligned} U_{xy} &= (U_v + 4U_\omega)_v v_y + (U_v + 4U_\omega)_\omega \omega_y \\ &= -(U_{vv} + 4U_{\omega v}) - (U_{v\omega} + 4U_{\omega\omega}) \\ &= -U_{vv} - 5U_{v\omega} - 4U_{\omega\omega} \end{aligned}$$

(+1)
$$\begin{aligned} \text{Now } U_y &= U_v v_y + U_\omega \omega_y = -U_v - U_\omega \\ U_{yy} &= (-U_v - U_\omega)_v v_y + (-U_v - U_\omega)_\omega \omega_y \\ &= U_{vv} + U_{\omega v} + U_{v\omega} + U_{\omega\omega} \\ &= U_{vv} + 2U_{v\omega} + U_{\omega\omega} \end{aligned}$$

(+1)
$$\begin{aligned} \text{Now the equation } U_{xx} + 5U_{xy} + 4U_{yy} &= 0 \\ \text{becomes } \cancel{U_{vv}} + 8U_{v\omega} + 16\cancel{U_{\omega\omega}} - 5\cancel{U_{vv}} - 25U_{v\omega} - 20\cancel{U_{\omega\omega}} \\ + 4\cancel{U_{vv}} + 8U_{v\omega} + 4\cancel{U_{\omega\omega}} &= 0 \\ \Rightarrow -9U_{v\omega} = 0 &\Rightarrow \boxed{U_{v\omega} = 0} \end{aligned}$$

(d) The normal form is $U_{vw} = 0$

Integrating with respect to w ,
 we get $U_v = f(v)$ (a constant with respect to v)

Now integrating with respect to v ,

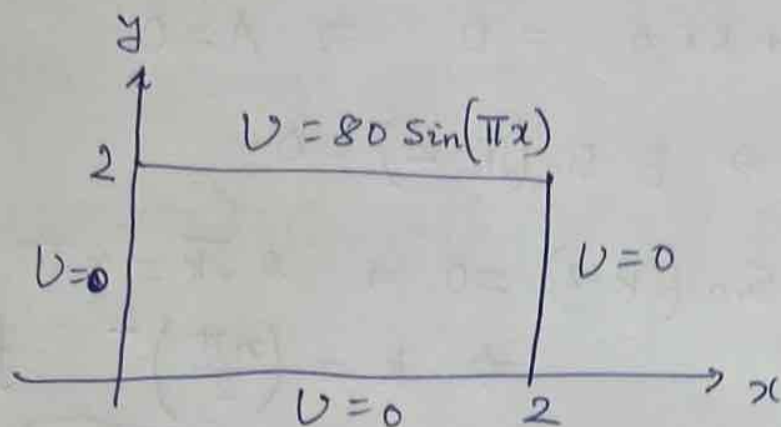
$$U = \int f(v) dv$$

+2 $\Rightarrow U = f_1(v) + f_2(w)$ (a constant with respect to w)

$\Rightarrow \boxed{U = f_1(x-y) + f_2(4x-y)}$ (This is the general solution)

(4)

(12)



Let $V(x, y)$ be the steady state temperature.

We need to solve $V_{xx} + V_{yy} = 0$ for the above square with the given boundary condition

Let us solve using separation of variable.

Let $V(x, y) = F(x)G(y)$

Then the equation becomes $F_{xx}G + FG_{yy} = 0$

$$\Rightarrow \frac{F_{xx}}{F} = - \frac{G_{yy}}{G} = \text{Constant} = -k \text{ (say)}$$

$\Rightarrow \left. \begin{aligned} F_{xx} + kF &= 0 \\ G_{yy} - kG &= 0 \end{aligned} \right\}$

Now note that if k is negative or zero, we will not get nontrivial solution.

Hence $F = A \cos(\sqrt{k}x) + B \sin(\sqrt{k}x)$

Now from the boundary condition $F(0)=0$, $F(2)=0$

$$\Rightarrow A \times 1 + B \times 0 = 0 \Rightarrow A = 0$$

(+4) Now $F(2) = 0 \Rightarrow B \sin(\sqrt{k} \cdot 2) = 0$

For nontrivial solution B to

$$\Rightarrow \sin(\sqrt{k} \cdot 2) = 0 \Rightarrow 2\sqrt{k} = n\pi$$

$$\Rightarrow k = \left(\frac{n\pi}{2}\right)^2 \text{ for } n = 1, 2, \dots$$

So, non-zero solutions are,

$$F(x) = \sin\left(\frac{n\pi}{2} x\right) \text{ for } n = 0, 1, 2, \dots$$

Now $G_{yy} - kG = 0$ and $k = \left(\frac{n\pi}{2}\right)^2$

$$\Rightarrow G_{yy} - \left(\frac{n\pi}{2}\right)^2 G = 0$$

Hence $G(y) = A_n e^{\frac{n\pi}{2} y} + B_n e^{-\frac{n\pi}{2} y}$

Now $G(0) = 0 \Rightarrow A_n + B_n = 0$
 $\Rightarrow B_n = -A_n$

Then $G(y) = A_n e^{\frac{n\pi}{2} y} - A_n e^{-\frac{n\pi}{2} y}$
 $= A_n \left(e^{\frac{n\pi}{2} y} - e^{-\frac{n\pi}{2} y} \right)$

$$= 2 A_n \sin\left[h \left(\frac{n\pi}{2} y\right)\right]$$

$$= C_n \sin\left[h \left(\frac{n\pi}{2} y\right)\right] \text{ where } C_n = 2A_n.$$

(+4)

Thus the general solution is

$$V(x, y) = \sum_{n=1}^{\infty} C_n \sin \left[h \left(\frac{n\pi}{2} y \right) \right] \sin \left(\frac{n\pi}{2} x \right)$$

Now at $y=2$

$$V(x, 2) = \sum_{n=1}^{\infty} C_n \sin [h(n\pi)] \sin \left(\frac{n\pi}{2} x \right)$$

It is given that $V(x, 2) = 80 \sin(\pi x) = 80 \sin \left(\frac{2\pi}{2} x \right)$

So, Comparing , $C_2 \sin [h(2\pi)] = 80$
and $C_n = 0$ for $n \neq 2$ }

$$\Rightarrow C_n = 0 \text{ for } n \neq 2$$

and $C_2 = \frac{80}{\sin [h(2\pi)]}$

So, the solution is

$$V(x, y) = \frac{80}{\sin [h(2\pi)]} \sin [h(\pi y)] \sin(\pi x)$$

x2

5 $y'' + \omega^2 y = r(t), \quad |\omega| \neq 0, 1, 2, \dots$

$r(t) = |t| \quad (2\pi - \text{periodic}) \quad -\pi \leq t \leq \pi$

Since $|t|$ is even, its Fourier series will be a cosine series.

Thus $|t| = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nt)$

where $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} |t| dt = \frac{2}{\pi} \int_0^{\pi} t dt \quad (\text{since } |t| \text{ is even})$

$$= \frac{2}{\pi} \left[\frac{t^2}{2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{\pi^2}{2} = \pi$$

So, $\boxed{\frac{a_0}{2} = \frac{\pi}{2}}$

+2 $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} |t| \cos(nt) dt$

$$= \frac{2}{\pi} \int_0^{\pi} t \cos(nt) dt$$

Now $\int t \cos(nt) dt = \frac{t \sin(nt)}{n} - \int \frac{\sin(nt)}{n} dt$

$$= \frac{t \sin(nt)}{n} + \frac{\cos(nt)}{n^2}$$

Therefore

$$a_n = \frac{2}{\pi} \left[\frac{t \sin(nt)}{n} + \frac{\cos(nt)}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[0 + \frac{\cos(n\pi)}{n^2} - \left(0 + \frac{1}{n^2} \right) \right]$$

$$= \frac{2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right]$$

So, $a_n = 0$ if ~~_____~~ n is even
 $= \frac{2}{\pi} \left(-\frac{2}{n^2} \right) = -\frac{4}{\pi n^2}$ if n is odd.

Therefore $|t| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{\substack{n=1 \\ (n \text{ odd})}}^{\infty} \frac{\cos nt}{n^2}$ ----- (1)

Now since the RHS of the differential equation has no sine terms, we will assume the solution is

$$y(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(nt)$$

$$y'(t) = \sum_{n=1}^{\infty} -n A_n \sin(nt), \quad y''(t) = \sum_{n=1}^{\infty} (-n^2) A_n \cos(nt)$$

So, $y'' + \omega^2 y = \sum_{n=1}^{\infty} (-n^2) A_n \cos(nt) + \frac{\omega^2 A_0}{2} + \omega^2 \sum_{n=1}^{\infty} A_n \cos(nt)$
 $= \frac{\omega^2 A_0}{2} + \sum_{n=1}^{\infty} (\omega^2 - n^2) A_n \cos(nt)$ ----- (2)

(+2)

(+4)

(16)

(17)

Comparing (1) and (2)

$$\frac{\omega^2 A_0}{2} = \frac{\pi}{2} \Rightarrow A_0 = \frac{\pi}{\omega^2}$$

$$(\omega^2 - n^2) A_n = \begin{cases} -\frac{4}{\pi n^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

$$\Rightarrow A_n = \begin{cases} \frac{-4}{\pi n^2 (\omega^2 - n^2)} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

Therefore the solution is

$$y(t) = \frac{\pi}{2\omega^2} - \frac{4}{\pi} \sum_{\substack{n=1 \\ (n \text{ odd})}}^{\infty} \frac{\cos(nt)}{n^2 (\omega^2 - n^2)}$$

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Question 6 :

Since $P_m(x)$ and $P_n(x)$ satisfy Legendre equation, we have

$$(1-x^2) P_m'' - 2x P_m' + m(m+1) P_m = 0 \quad \dots\dots (1)$$

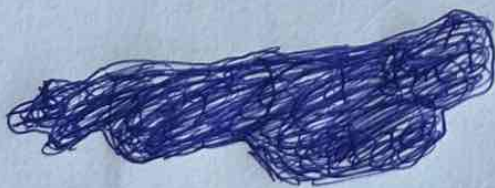
$$(1-x^2) P_n'' - 2x P_n' + n(n+1) P_n = 0 \quad \dots\dots (2)$$

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Now $(1) \times P_n - (2) \times P_m$ implies

$$\begin{aligned} & (1-x^2) P_m'' P_n - 2x P_m' P_n + m(m+1) P_m P_n \\ & - (1-x^2) P_n'' P_m + 2x P_n' P_m - n(n+1) P_n P_m = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow & (1-x^2) (P_m'' P_n - P_n'' P_m) - 2x (P_m' P_n - P_n' P_m) \\ & + [m(m+1) - n(n+1)] P_m P_n = 0 \end{aligned}$$



$$\begin{aligned} \Rightarrow & (1-x^2) (P_m'' P_n + P_m' P_n' - P_m' P_n' - P_n'' P_m) \\ & - 2x (P_m' P_n - P_n' P_m) + [m(m+1) - n(n+1)] P_m P_n = 0 \end{aligned}$$

$$\Rightarrow (1-x^2) \frac{d}{dx} (P_m' P_n - P_n' P_m) - 2x (P_m' P_n - P_n' P_m)$$

$$= [n(n+1) - m(m+1)] P_m P_n$$

$$\Rightarrow \frac{d}{dx} \left\{ (1-x^2) (P_m' P_n - P_m P_n') \right\} \\ = (n^2 + n - m^2 - m) P_n P_m$$

$$\Rightarrow \frac{d}{dx} \left\{ (1-x^2) (P_m' P_n - P_m P_n') \right\} = (n-m)(n+m+1) P_n P_m$$

Integrating both sides with respect to x from -1 to 1 we get

$$\left[(1-x^2) P_m' P_n - P_m P_n' \right]_{-1}^1 = \int_{-1}^1 (n-m)(n+m+1) P_n P_m dx$$

$$\Rightarrow (n-m)(n+m+1) \int_{-1}^1 P_n P_m dx = 0$$

$$\Rightarrow \int_{-1}^1 P_n P_m dx = 0 \quad \text{if } n \neq m$$

(7) Since all zeros of $Q(s)$ are distinct and the coefficient of the highest power of s is 1, we can write

$$Q(s) = (s - \alpha_1)(s - \alpha_2) \cdots (s - \alpha_n)$$

} +2

Now since degree of $P <$ degree of Q ,

the partial fraction decomposition of $\frac{P(s)}{Q(s)}$ can

be written as

$$\frac{P(s)}{Q(s)} = \frac{A_1}{(s - \alpha_1)} + \frac{A_2}{(s - \alpha_2)} + \cdots + \frac{A_n}{(s - \alpha_n)} = \sum_{k=1}^n \frac{A_k}{(s - \alpha_k)}$$

$$\Rightarrow P(s) = \frac{A_1 Q(s)}{(s - \alpha_1)} + \frac{A_2 Q(s)}{(s - \alpha_2)} + \cdots + \frac{A_n Q(s)}{(s - \alpha_n)}$$

$$\Rightarrow P(s) = A_1 (s - \alpha_2) \cdots (s - \alpha_n) + A_2 (s - \alpha_1)(s - \alpha_3) \cdots (s - \alpha_n) + \cdots + A_n (s - \alpha_1) \cdots (s - \alpha_{n-1})$$

Substituting $s = \alpha_k$ we get

$$P(\alpha_k) = A_k (\alpha_k - \alpha_1) \cdots (\alpha_k - \alpha_{k-1})(\alpha_k - \alpha_{k+1}) \cdots (\alpha_k - \alpha_n)$$

for $k = 1, 2, \dots, n$

$$\Rightarrow P(\alpha_k) = A_k Q'(\alpha_k)$$

$$\Rightarrow A_k = \frac{P(\alpha_k)}{Q'(\alpha_k)} \quad \text{for } k = 1, 2, \dots, n$$

} +2

} +6

Hence
$$\frac{P(s)}{Q(s)} = \sum_{k=1}^n \frac{A_k}{(s-\alpha_k)} = \sum_{k=1}^n \frac{P(\alpha_k)}{Q'(\alpha_k)} \frac{1}{(s-\alpha_k)}$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\} = \sum_{k=1}^n \frac{P(\alpha_k)}{Q'(\alpha_k)} \mathcal{L}^{-1} \left\{ \frac{1}{(s-\alpha_k)} \right\}$$

+ 2

$$\Rightarrow \boxed{\mathcal{L}^{-1} \left\{ \frac{P(s)}{Q(s)} \right\} = \sum_{k=1}^n \frac{P(\alpha_k)}{Q'(\alpha_k)} e^{\alpha_k t}}$$